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Module Objectives
Electronic Codebook (ECB) Mode
Cipher Block Chaining (CBC) Mode
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Block Cipher Operation Modes
Reading: Lecture Slides for Block Cipher Operation Modes
15 min
Quiz: Block Cipher Operation Modes
40 min
Real-World Use of Block Cipher Operation Mode

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Block Cipher Operation Modes

Review Learning Objectives

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Due Jun 9, 11:59 PM IST Attempts 3 every 8 hours
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1. Which of the following describes the 2nd block encryption output of CBC mode (C2). The followings are given: E (an encryption cipher function), K (the key). C1 (the cipher output from the 1st block encryption), P2 (the plaintext for the 2nd block), and XOR (and exclusive-or function). For example, XOR(x,y) indicate the bit-by-bit XOR between the two bit vectors x and y, and E(K,P2) is the encryption output when using the key K and the plaintext P2. **3 / 3 points**

- ○ C2 = XOR(P2,E(K,XOR(C1,P1)))
- ○ C2 = E(K,XOR(C1,P1))
- ○ C2 = XOR(P1,E(K,XOR(C1,P1)))
- ○ C2 = XOR(P2,E(K,XOR(C1,P2)))
- ◉ C2 = E(K,XOR(C1,P2))
- ○ C2 = XOR(E(C1),K)

✓ **Correct**
See the CBC Mode lesson.

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2. Which of the following operation modes do not require padding (if the data does not fill the block)? Check all that applies. **3 / 3 points**

- ☑ CTR

✓ **Correct**

- ☐ CBC
- ☐ ECB
- ☑ CFB

✓ **Correct**

- ☑ OFB

✓ **Correct**

3. Which of the followings describe the 2nd block encryption output (C2) of OFB mode? The followings are given: E (an encryption cipher function), K (the key). C1 (the cipher output from the 1st block encryption), P2 (the plaintext for the 2nd block), P1 (the plaintext for the 1st block), and XOR (and exclusive-or function). For example, XOR(x,y) indicate the bit-by-bit XOR between the two bit vectors x and y, and E(K,P1) is the encryption output when using the key K and the plaintext P1. **3 / 3 points**

- ◉ C2 = XOR(P2,E(K,XOR(C1,P1)))
- ○ C2 = XOR(P2,E(K,XOR(C1,P2)))
- ○ C2 = E(K,XOR(C1,P2))
- ○ C2 = XOR(P1,E(K,XOR(C1,P1)))
- ○ C2 = XOR(E(C1),K)
- ○ C2 = E(K,XOR(C1,P1))

✓ **Correct**
See the OFB Mode lesson.

4. Suppose a small, low-power device experiences that a real-time computation of the encryption/decryption function is burdensome (e.g., serving as the bottleneck of the performance) and wants to compute the encryption/decryption functions offline (e.g., computed before the plaintext/ciphertext). Which of the following block cipher operation modes support such feature? Check all that applies. **2.25 / 3 points**

- ☐ CBC
- ☐ OFB
- ☐ CFB
- ☑ CTR

✓ **Correct**
This is described in the CTR Mode Lecture

You didn't select all the correct answers

5. You are given a block cipher with a block size of 64 bits. Given a data input of 32 bits, how many bits do you need to pad if the using the ECB operation mode? **3 / 3 points**

32

✓ **Correct**
Please see the ECB Mode Lesson.

6. You are given a block cipher with a block size of 64 bits. Given a data input of 32 bits, how many bits do you need to pad if the using the CFB operation mode? **3 / 3 points**

0

✓ **Correct**
Please see the OFB Mode Lesson.

7. Alice and Bob is not synchronized in time, and their clock frequency is different. Which of the block cipher operation modes would be affected by such lack of synchronization? Check all that applies. **3 / 3 points**

- ☑ CTR

✓ **Correct**
As has been reviewed in the CTR Mode lecture, CTR mode is dependent on time.

- ☐ OFB
- ☐ CBC
- ☐ CFB
- ☐ ECB

8. Suppose Pi indicates the plaintext input for the block cipher i from the sender's perspective, e.g., P2 is the plaintext input for the 2nd block cipher, and Ci is the ciphertext input for the block cipher i from the receiver's perspective. (If there were no errors, Ci is also the ciphertext output of the block cipher i from the sender.) Which of the following scenarios causing errors affect C3 (and cause error in C3)? Select all that applies. **4.285714285714286 / 6 points**

- ☐ Error occurs on C2 for CTR Mode.
- ☐ Error occurs on C2 for ECB Mode.
- ☑ Error occurs on P2 for CFB Mode.

✓ **Correct**

- ☐ Error occurs on C1 for CBC Mode.
- ☐ Error occurs on C2 for OFB Mode.
- ☐ Error occurs on P1 for CBC Mode.
- ☑ Error occurs on C2 for CFB Mode.

⊗ **This should not be selected**